



Data Warehouse

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Objectives

- x Explain Data Warehouse concept
 - x Describe Data Warehouse Architecture
 - x Explain the Data Warehouse Tools and Techniques
 - x Explain Data Marts and Data Mining
 - x Designing a Data Warehousing Database
 - x Designing a Dimensionality Modeling
 - x Designing the Multidimensional Data Representation
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Introduction to Data Warehouse

- x A data warehouse is a collection of integrated databases designed to support a Decision Support System.
 - x A central data repository where data from operational database and other sources are integrated, cleaned, and standardized to support decision making.
 - x Objective of DW is to evaluate future strategy of a company.
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x For example, it increased Capital One's customers from 1 million to approximately 9 millions in 8 years.

x Just like a muscle: DW increases in strength with active use.

x With each new test and product, valuable information is added to the DW, allowing the analyst to learn from the success and failure of the past.

x The key to survival:

x Is the ability to analyze, plan, and react to changing business conditions in a much more rapid fashion.



Which are our lowest/highest margin customers ?

Who are my customers and what products are they buying?

What is the most effective distribution channel?

What product promotions have the biggest impact on revenue?

Which customers are most likely to go to the competition ?



Characteristic of Data Warehouse

Subject-oriented

Can be used to analyze a particular subject area.

Integrated

Integrates data from multiple data sources.

Time-variant

Historical data is kept in a data warehouse.

Non-volatile

Once data is in the data warehouse, it will not change.

DW VS OPERATIONAL DB (OLTP)

Characteristic	Operational Data Store	Data Warehouse
How is it built?	One application or subject area at a time.	Typically multiple subject areas at a time
Area of support?	Day-to-day business operations.	Decision support for managerial activities.
Currency of data?	Up-to-the-minute, real time.	Typically represents a static point in time.
Typical unit for analysis?	Small, manageable, transaction level units.	Large, unpredictable, variable units.
Design focus?	High-performance, limited flexibility.	High flexibility, high performance.

To summarize ...

- OLTP Systems are used to “*run*” a business



- The Data Warehouse helps to “*optimize*” the business

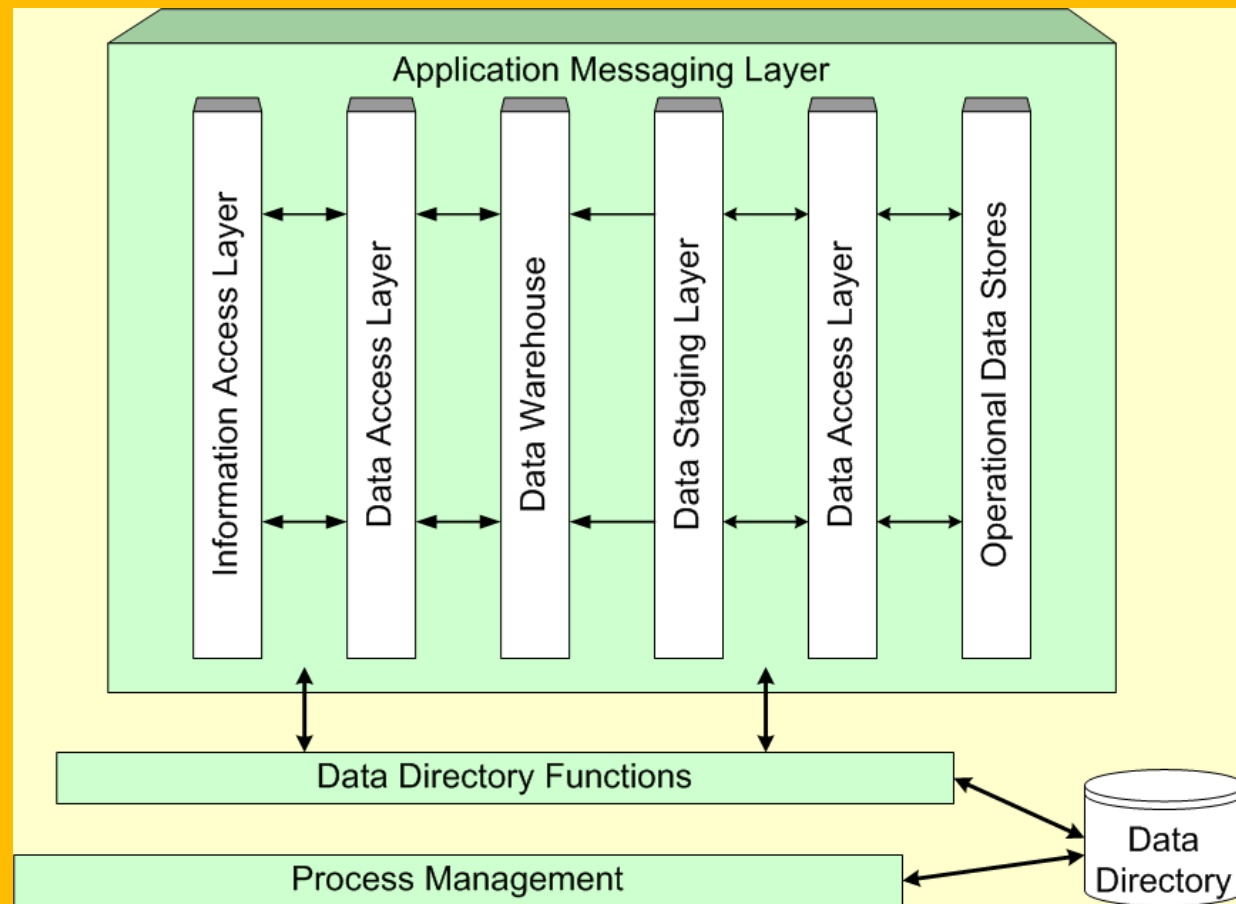


Data Warehouse Architecture

The architecture consists of various interconnected elements:

- x *Operational and external database layer* – the source data for the DW
 - x *Information access layer* – the tools the end user access to extract and analyze the data
 - x *Data access layer* – the interface between the operational and information access layers
 - x *Metadata layer* – the data directory or repository of metadata information
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Components of the Data Warehouse Architecture

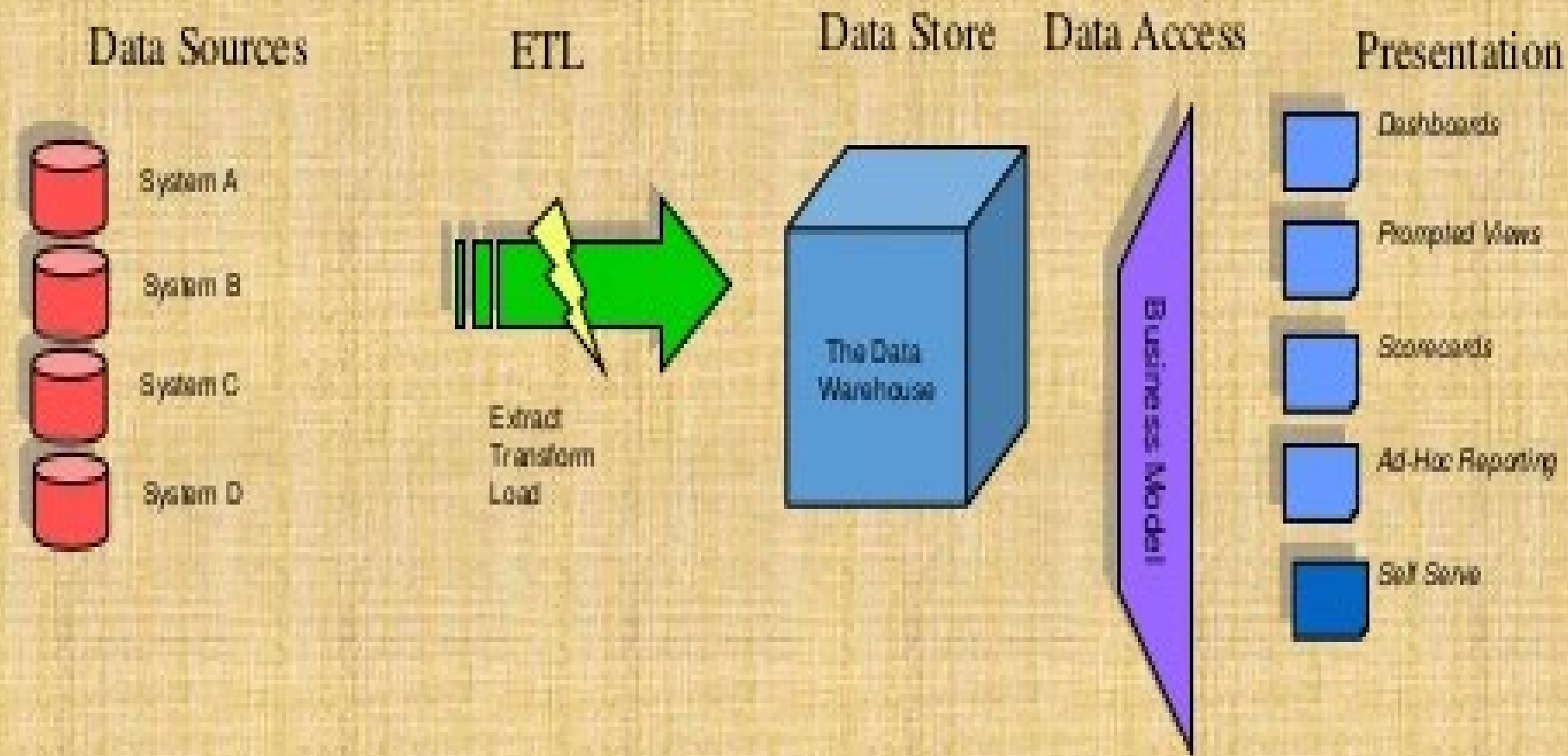




Additional layers are:

- X *Process management layer* – the scheduler or job controller
 - X *Application messaging layer* – the “middleware” that transports information around the firm
 - X *Physical data warehouse layer* – where the actual data used in the DSS are located
 - X *Data staging layer* – all of the processes necessary to select, edit, summarize and load warehouse data from the operational and external data bases
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Typical DW Architecture





Data Warehouse Tools and Techniques

- X ETL Process

- X Extract – take the data from operational DB or any reliable sources to be integrated.
 - X Transform – change the data format such as data merging, calculation and data summarization. But before transform, it must be cleaned to free from any data error.
 - X Loading – once the data is summarized, it must be loaded into the DW.
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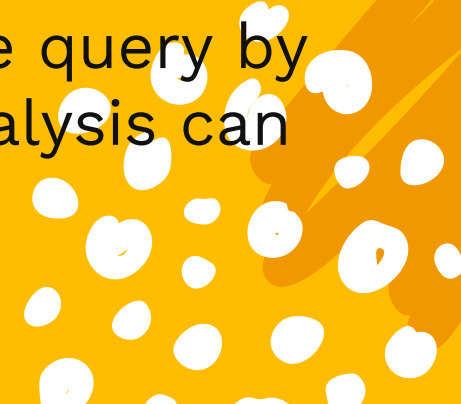


DW DBMS

“ Choose the right software to run a data warehouse.

“ Requirements needed:

- x Load Performance – must be fast as the data is huge.
 - x Load Processing – load data into a data warehouse must be done periodically and only involve changed data.
 - x Data Quality Management – must give user precise result eventhough data is huge.
 - x Query Performance – ad hoc analysis must not be slow.
 - x Terabyte scalability – size can be scale up when needed.
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- x Requirements needed:
 - x Mass user scalability – number of users can be increased without affecting the performance.
 - x Networked DW – should be able to cooperating in larger network environment.
 - x Warehouse administration – the DBA must be ready all the time to support any operation in the warehouse.
 - x Integrated dimensional analysis – data representation can be done from the DW.
 - x Advanced Query Functionality – any advance query by the user for calculation and comparative analysis can be done.



x Data Warehouse Metadata:

- ❖ For example, a line in a sales database may contain:

4056 KJ596 223.45

- ❖ This is mostly meaningless until we consult the metadata that tells us it was store number 4056, product KJ596 and sales of \$223.45
 - ❖ The metadata are essential ingredients in the transformation of raw data into knowledge. They are the “keys” that allow us to handle the raw data.
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General metadata issues associated with Data Warehouse use:

- x What tables, attributes and keys does the DW contain?
 - x Where did each set of data come from?
 - x What transformations were applied with cleansing?
 - x How have the metadata changed over time?
 - x How often do the data get reloaded?
 - x Are there so many data elements that you need to be careful what you ask for?
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x Component of DW Metadata

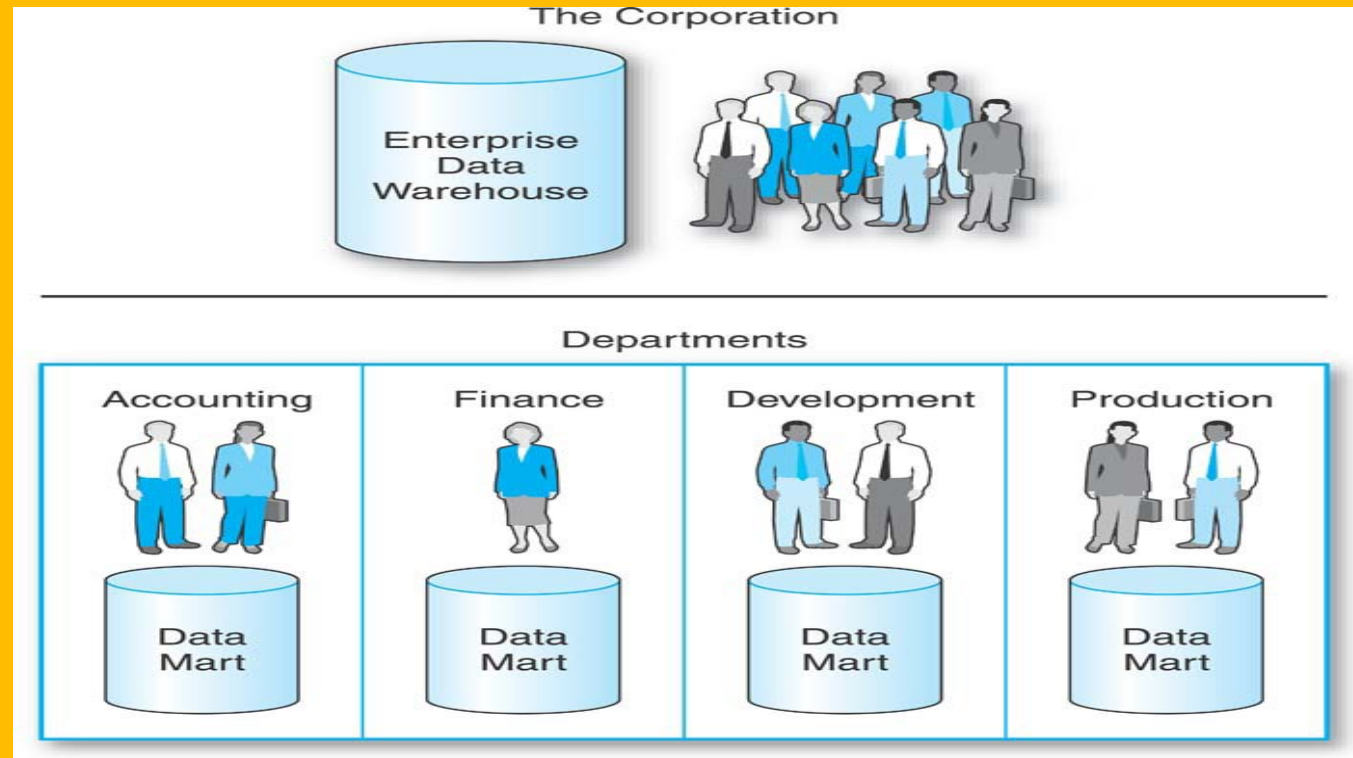
- x Transformation maps – records that show what transformations were applied
 - x Extraction & relationship history – records that show what data was analyzed
 - x Algorithms for summarization – methods available for aggregating and summarizing
 - x Data ownership – records that show origin
 - x Patterns of access – records that show what data are accessed and how often
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Example of DW Applications

Industry	Key Applications
Airline	Yield management, route assessment
Telecommunications	Customer retention, network design
Insurance	Risk assessment, product design, fraud detection
Retail	Target marketing, supply-chain management

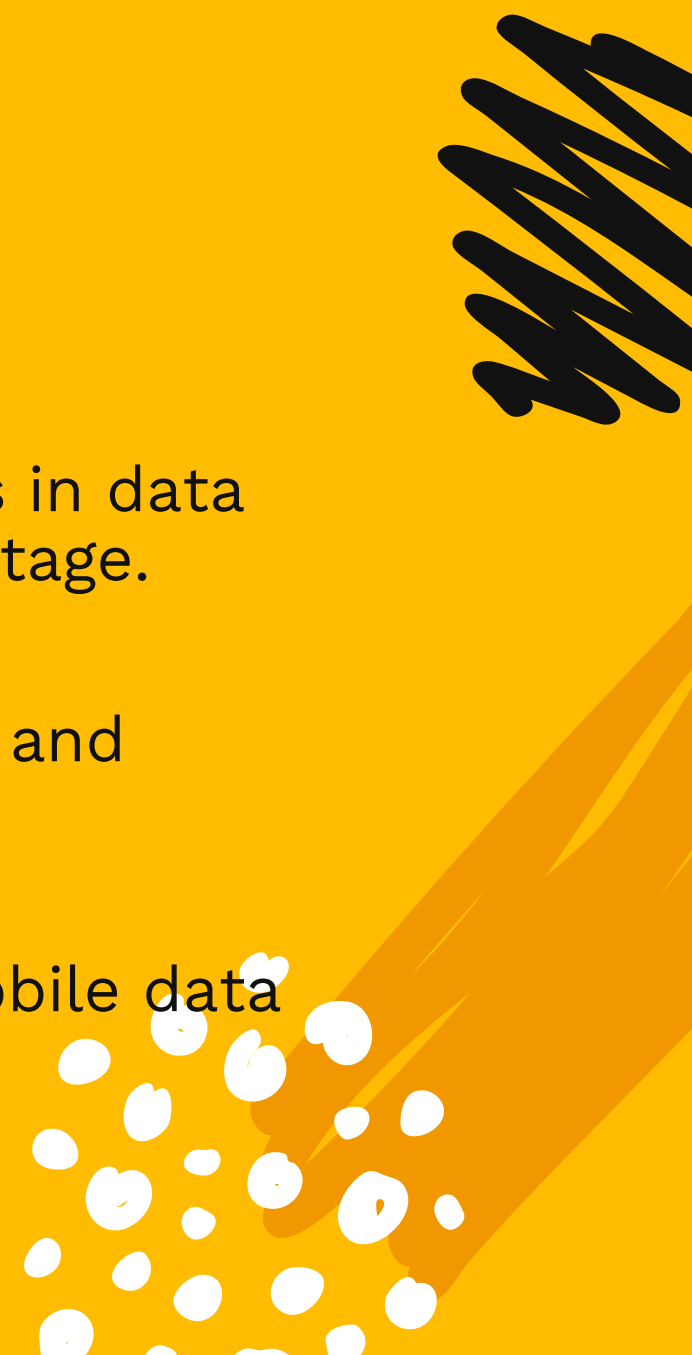
Data Marts

- x It is a subset or view of a data warehouse
- x Typically, at a department or functional level.





Data Mining

- x The process of discovering implicit patterns in data and using these patterns for business advantage.
 - x Facilitates the ability to detect, understand, and predict patterns.
 - x Example of data mining application is on mobile data usage for telecommunication industry.
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Designing a Data Warehousing Database

Designing a DW is highly complex. These questions must be answered before start designing:

1. Which user requirements are most important?
2. Which data should be considered first?
3. Should the project be scaled down?

Why? Because a failed DW project could lead to company bankruptcy.





Methodology

- X There are 2 different methodologies suggested by Inmon and Kimball.
 - X Both Kimball and Inmon's architectures share a same common feature that each has a single integrated repository of atomic data.
 - X In Inmon's architecture, it is called *enterprise data warehouse*. And in Kimball's architecture, it is known as the *dimensional data warehouse*.
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- x Both architectures have an enterprise focus that supports information analysis across the organization.
 - x This approach enables to address the business requirements not only within a subject area but also across subject areas.
 - x However there are some differences in the data warehouse architectures of both experts:

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1. Kimball uses the dimensional model such as star schemas or snowflakes to organize the data in *dimensional data warehouse* while Inmon uses ER model in enterprise data warehouse.

Inmon only uses dimensional model for data marts only while Kimball uses it for all data



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2. Inmon uses data marts as physical separation from enterprise data warehouse and they are built for departmental uses.

While in Kimball's architecture, it is unnecessary to separate the data marts from the dimensional data warehouse.



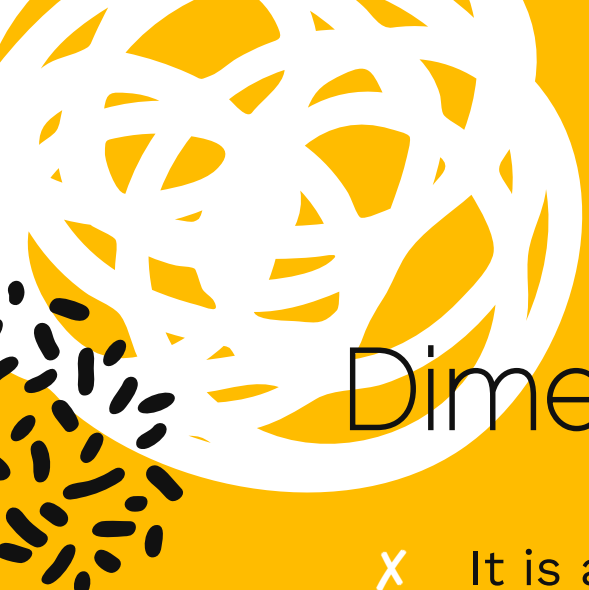


3. In dimensional data warehouse of Kimball, analytic systems can access data directly.

While in Inmon's architecture, analytic systems can only access data in enterprise data warehouse via data marts.



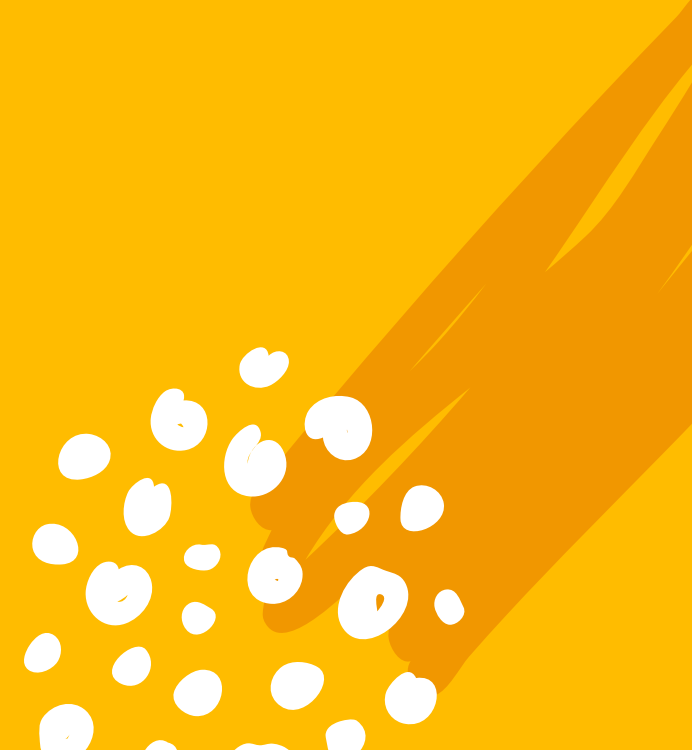
Characteristics	Favours Kimball	Favours Inmon
Business decision support requirements	Tactical	Strategic
Data integration requirements	Individual business requirements	Enterprise-wide integration
The structure of data	KPI, business performance measures, scorecards...	Data that meet multiple and varied information needs and non-metric data
Persistence of data in source systems	Source systems are quite stable	Source systems have high rate of change
Skill sets	Small team of generalists	Bigger team of specialists
Time constraint	Urgent needs for the first data warehouse	Longer time is allowed to meet business' needs.
Cost to build	Low start-up cost	High start-up costs



Dimensionality Modeling

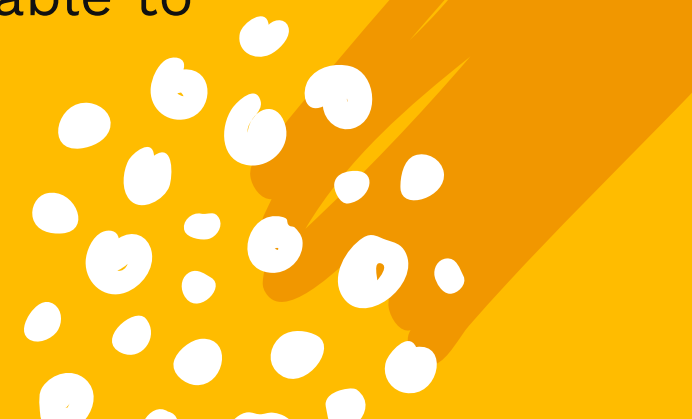
- X It is a logical design technique that aims to represent the data in standard, intuitive form that allows for high-performance access.

 - X It will look like ERD, but it can be called as either:
 1. Star Schema
 2. Snowflake Schema
 3. Constellation Schema

 - X Each schema must contain:
 1. Fact table – numeric table
 2. Dimension table – descriptive table
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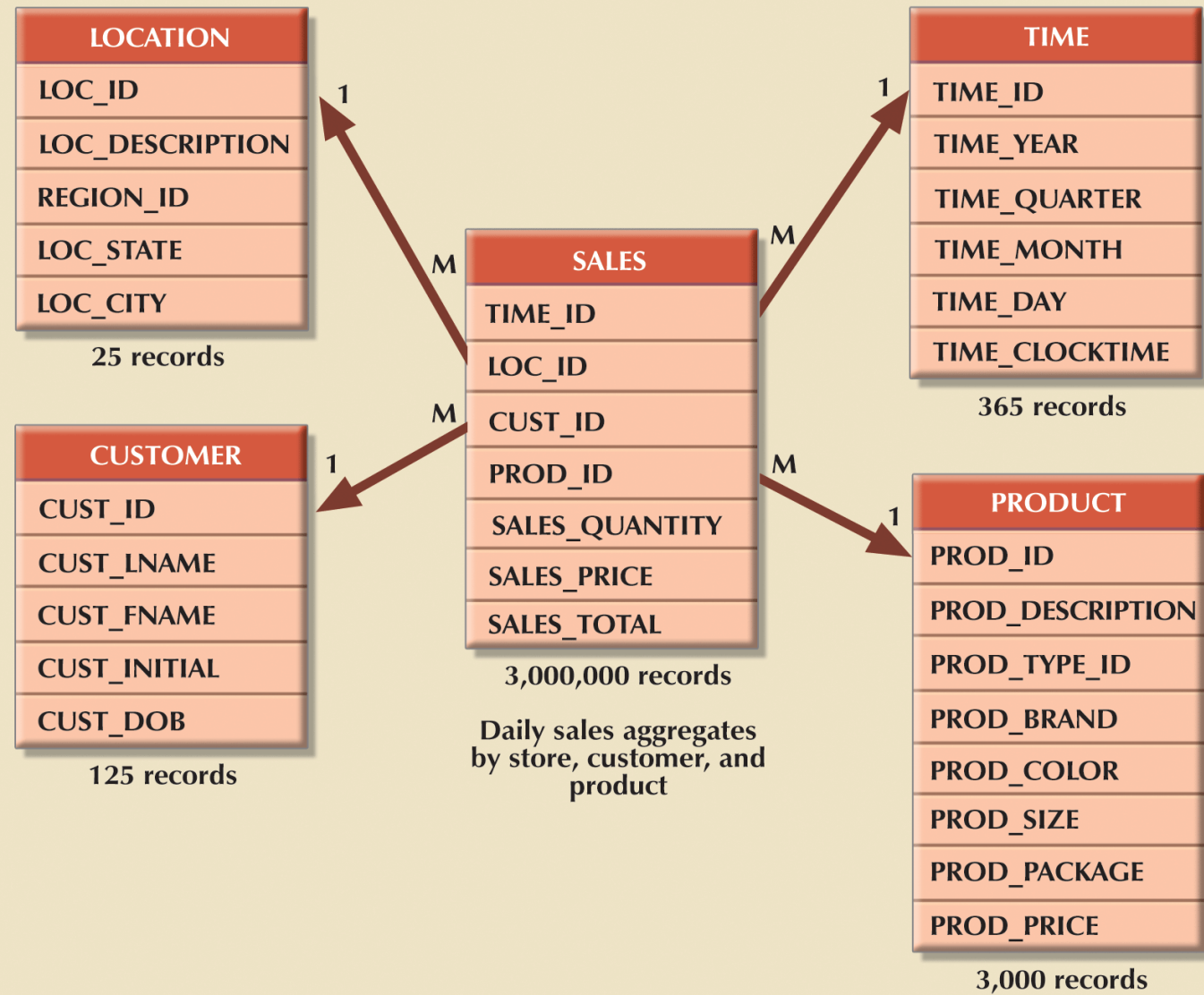


Star Schema

- x Data modeling representation of multidimensional database
 - x Star schema diagram looks like star with one large central table is fact table while other is dimension tables
 - x There is 1-M relationship from each dimension table to facts table
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**FIGURE
13.17**

Star schema for SALES



Product

<u>Product _Code</u>	Description	Color	Size
100	Sweater	Blue	40
110	Shoes	Brown	10 1/2
125	Gloves	Tan	M
• • •			

Period

<u>Period _Code</u>	Year	Quarter	Month
001	2004	1	4
002	2004	1	5
003	2004	1	6
• • •			

Sales

<u>Product _Code</u>	<u>Period _Code</u>	<u>Store _Code</u>	Units _Sold	Dollars _Sold	Dollars _Cost
110	002	S1	30	1500	1200
125	003	S2	50	1000	600
100	001	S1	40	1600	1000
110	002	S3	40	2000	1200
100	003	S2	30	1200	750
• • •					

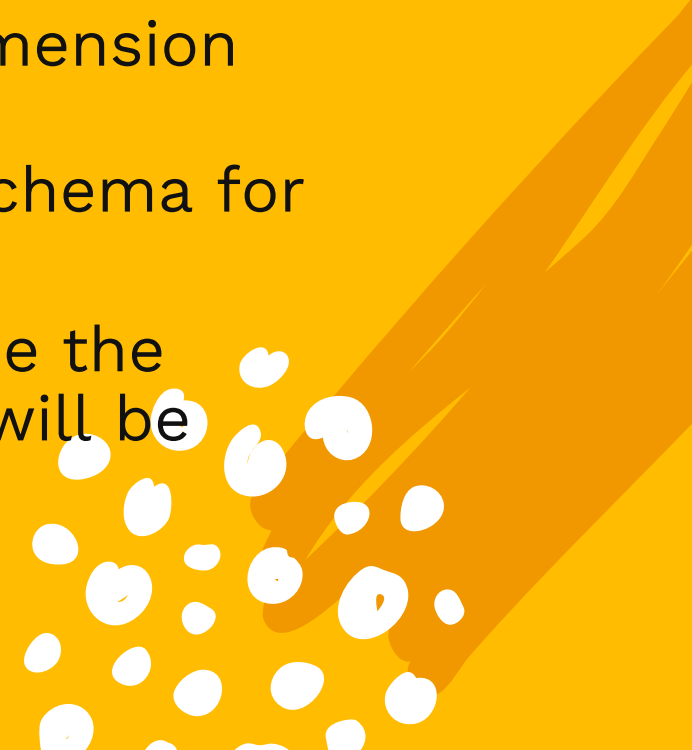
Store

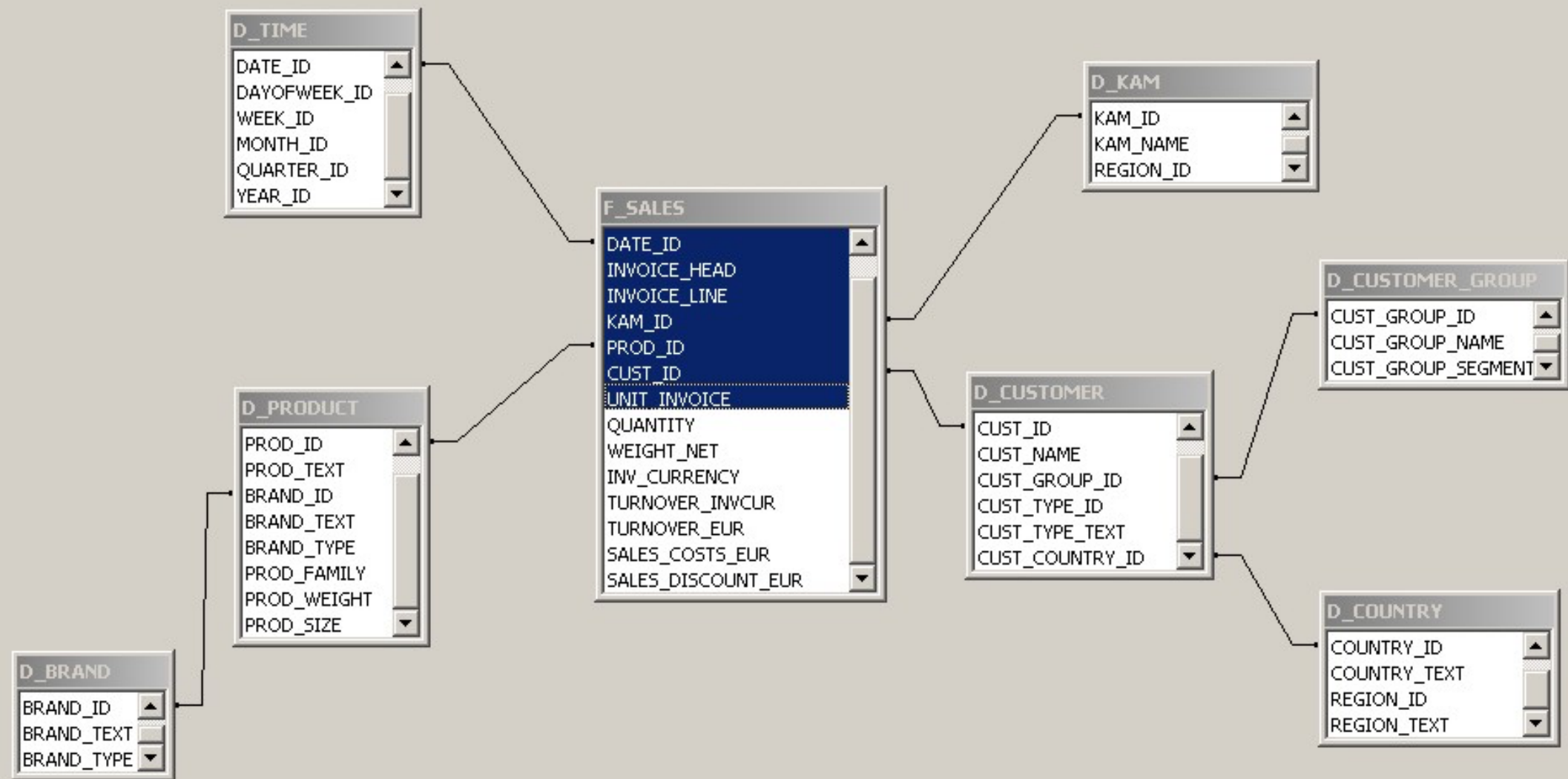
<u>Store _Code</u>	Store _Name	City	Telephone	Manager
S1	Jan's	San Antonio	683-192-1400	Burgess
S2	Bill's	Portland	943-681-2135	Thomas
S3	Ed's	Boulder	417-196-8037	Perry
• • •				



Snowflake Schema

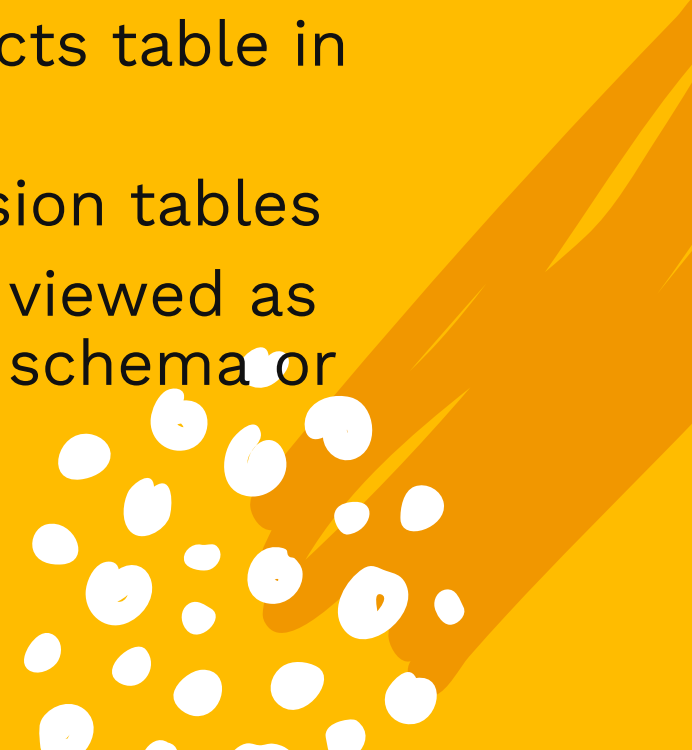


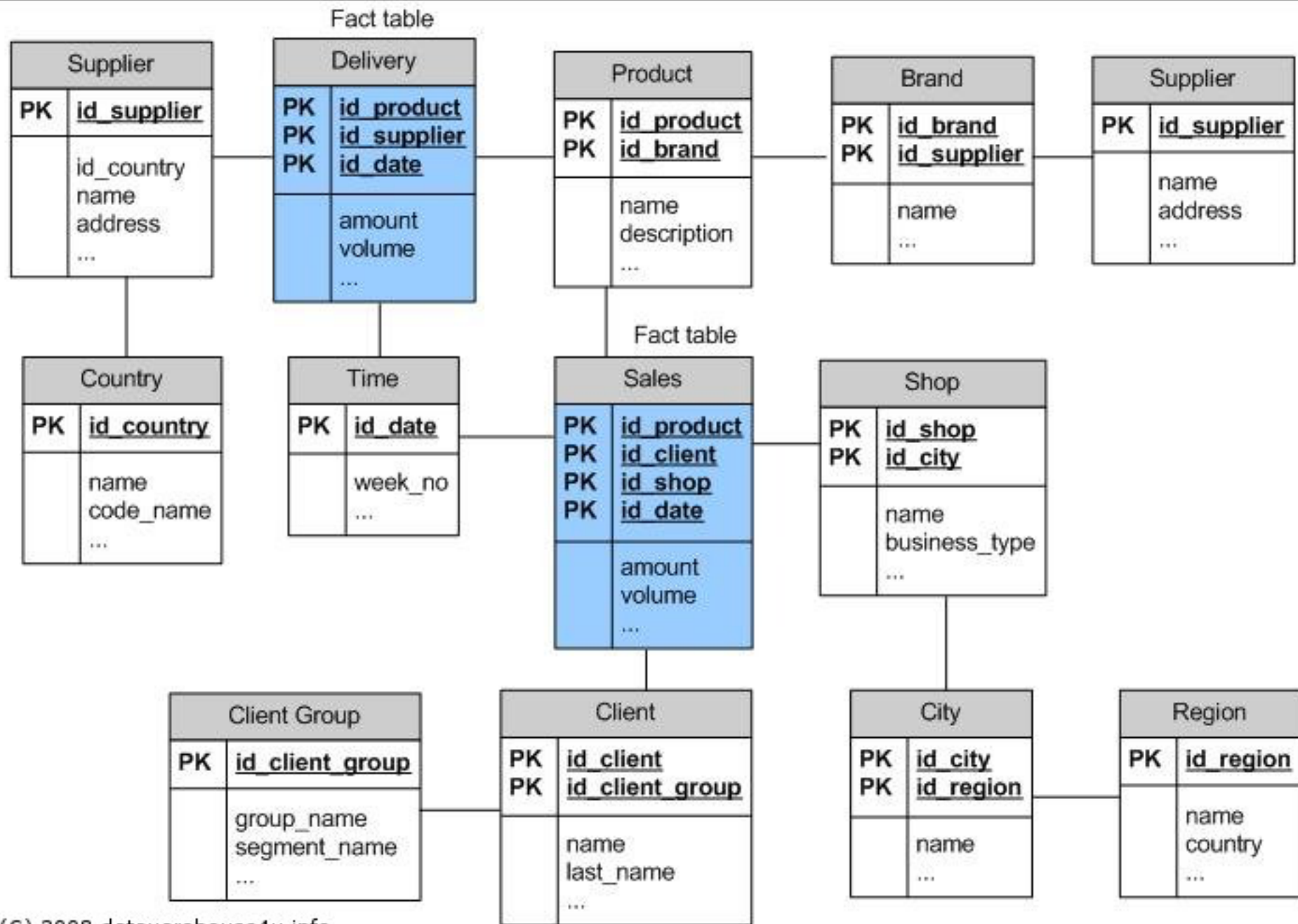
- x Data modeling representation of multidimensional database
 - x Snowflake schema has multiple levels of dimension tables related to one or more facts tables
 - x The snowflake schema instead of the star schema for small dimension tables that are not in 3NF
 - x However, the snowflake structure can reduce the effectiveness of browsing, since more joins will be needed
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Constellation Schema

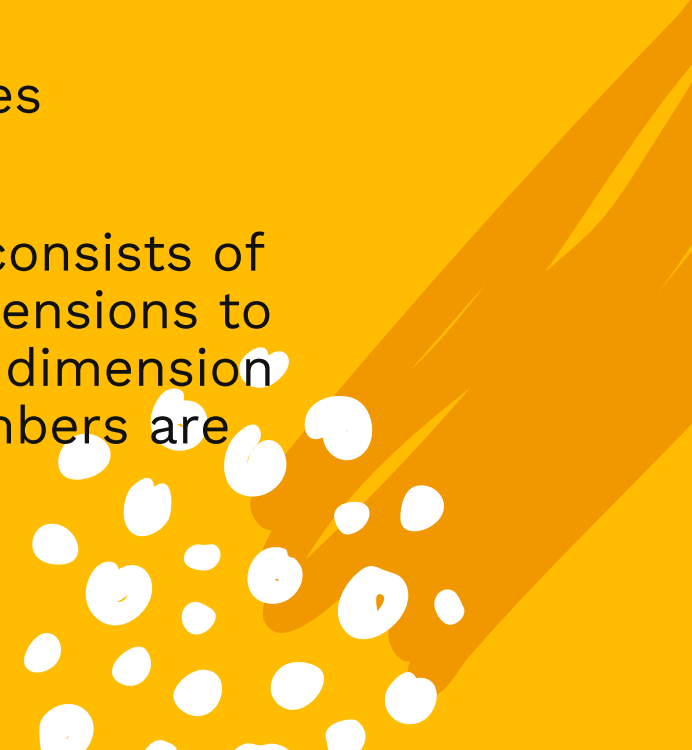
- x Data modeling representation of multidimensional database
 - x A constellation schema contains multiple facts table in the center related to the dimension table
 - x Typically, the facts table share some dimension tables
 - x Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation
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Year	Region	Agent	Product	Quantity
2009	East	Carlos	Erasers	50
2009	East	Tere	Erasers	12
2009	North	Carlos	Widgets	120
2009	North	Tere	Widgets	100
2009	North	Carlos	Widgets	30
2009	South	Victor	Balls	145
2009	South	Victor	Balls	34
2009	South	Victor	Balls	80
2009	West	Mary	Pencils	89
2009	West	Mary	Pencils	56
2010	East	Carlos	Pencils	45
2010	East	Victor	Balls	55
2010	North	Mary	Pencils	60
2010	North	Victor	Erasers	20
2010	South	Carlos	Widgets	30
2010	South	Mary	Widgets	75
2010	South	Mary	Widgets	50
2010	South	Tere	Balls	70
2010	South	Tere	Erasers	90
2010	West	Carlos	Widgets	25
2010	West	Tere	Balls	100



Multidimensional Data Representation

- X Multidimensional data model supports data representation and operations for decision support processing in data warehouse
 - X Users think about decision support data as data cubes
 - X Data cube or hypercube is multidimensional format consists of cells containing measures (eg sales amount) and dimensions to label numeric data (eg product, location, time). Each dimension contains values known as members (eg location members are California, Utah etc)
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PRODUCT	LOCATION	SALES
Mono Laser	California	80
Mono Laser	Utah	40
Mono Laser	Arizona	70
Mono Laser	Washington	75
Mono Laser	Colorado	65
Ink Jet	California	110
Ink Jet	Utah	90
Ink Jet	Arizona	55
Ink Jet	Washington	85
Ink Jet	Colorado	45
Photo	California	60
Photo	Utah	50
Photo	Arizona	60
Photo	Washington	45
Photo	Colorado	85
Portable	California	25
Portable	Utah	30
Portable	Arizona	35
Portable	Washington	45
Portable	Colorado	60

Location	Product			
	Mono Laser	Ink Jet	Photo	Portable
California	80	110	60	25
Utah	40	90	50	30
Arizon	70	55	60	35
Washington	75	85	45	45
Colorado	65	45	85	60

LOCATION

California

Utah

Arizona

Washington

Colorado

80

110

60

25

40

90

50

30

70

55

60

35

75

85

45

45

65

45

85

60

Mono
Laser

Ink
Jet

Photo

Portable

PRODUCT

31/12/2006

TIME

1/1/2006



Multidimensional Terminology:

- x Dimension: subject label for a row or column
 - x Member: value of dimension
 - x Measure: quantitative data stored in cells
 - x Sparsity: - large empty cells in a data cube.
 - Waste space and be slow to process.
- 

We have a multidimensional data model with the fact table **Sales** and the dimension tables **Customers**, **Products**, and **Salespeople**. The sales table below represents sales data for 1st January 2010:

CUSTID	PRODID	STAFFID	QUANTITY
101	1	10	20
101	2	11	10
101	3	10	5
102	2	11	8
102	3	10	24
103	1	11	50
103	2	11	45
103	3	10	30

- x Draw a 3D picture of a data cube. Assume that all values that are missing from the **Sales** table are 0.

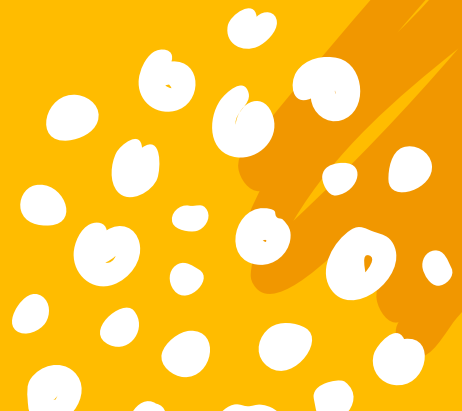


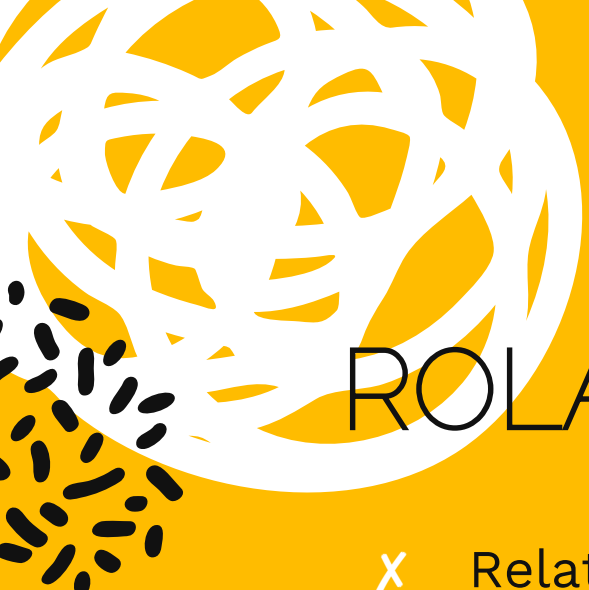
OLAP Servers

- X Online Analytical Processing Server (OLAP) is based on the multidimensional data model.
 - X It allows managers, and analysts to get an insight of the information through fast, consistent, and interactive access to information.
 - X This part will cover the types of OLAP, operations on OLAP, difference between OLAP, and statistical databases and OLTP.
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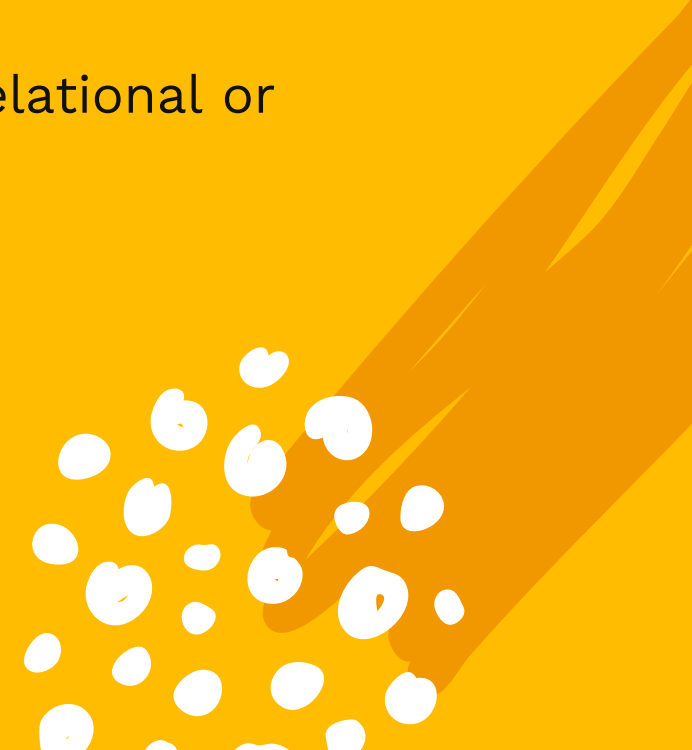


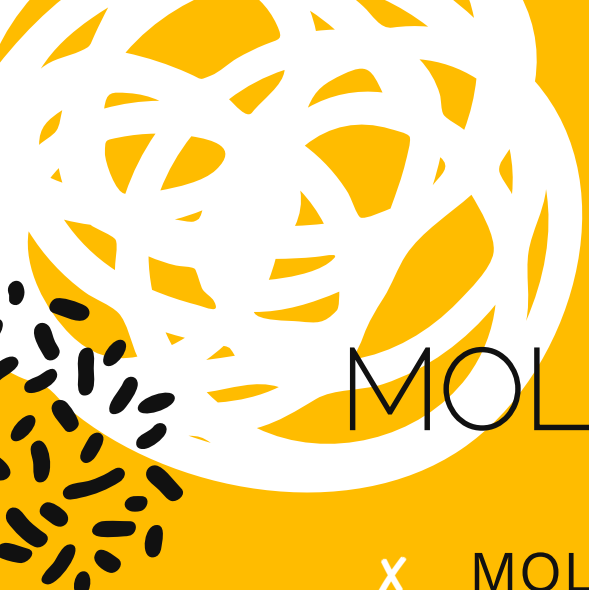
x There are four types of OLAP servers:

- x Relational OLAP (ROLAP)
 - x Multidimensional OLAP (MOLAP)
 - x Hybrid OLAP (HOLAP)
 - x Specialized SQL Servers (not covered)
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ROLAP (STAR SCHEMA)

- X Relational OLAP (ROLAP) servers are placed between relational back-end server and client front-end tools.
 - X To store and manage warehouse data, ROLAP uses relational or extended-relational DBMS.
 - X ROLAP includes the following:
 - X Implementation of aggregation navigation logic.
 - X Optimization for each DBMS back end.
 - X Additional tools and services.
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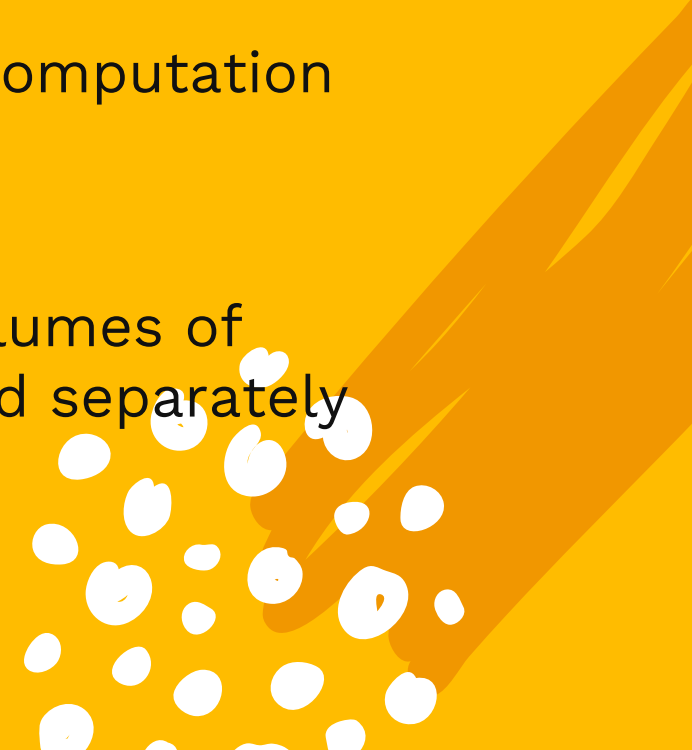


MOLAP (DATA CUBE)

- x MOLAP uses array-based multidimensional storage engines for multidimensional views of data.
 - x With multidimensional data stores, the storage utilization may be low if the data set is sparse.
 - x Therefore, many MOLAP server use two levels of data storage representation to handle dense and sparse data sets.
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HOLAP

- x Hybrid OLAP is a combination of both ROLAP and MOLAP.
 - x It offers higher scalability of ROLAP and faster computation of MOLAP.
 - x HOLAP servers allows to store the large data volumes of detailed information. The aggregations are stored separately in MOLAP store.
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Operator	Purpose	Description
Slice	Focus attention on a subset of dimensions	Replace a dimension with a single member value or with a summary of its measure values
Dice	Focus attention on a subset of member values	Replace a dimension with a subset of members
Drill-down	Obtain more detail about a dimension	Navigate from a more general level to a more specific level
Roll-up	Summarize details about a dimension	Navigate from a more specific level to a more general level
Pivot	Present data in a different order	Rearrange the dimensions in a data cube

SLICE OPERATOR

- Focus on a subset of dimensions
- Similar to restriction operator
- Dimension are set to specific value
- Set dimension to specific value: 1/1/2006

Example Slice Operation

Location	Product			
	Mono Laser	Ink Jet	Photo	Portable
California	80	110	60	25
Utah	40	90	50	30
Arizon	70	55	60	35
Washington	75	85	45	45
Colorado	65	45	85	60

Example Slice-Summarize Operation

Location	Time			Total Sales
	1/1/2006	1/2/2006		
California	80	110	...	16,250
Utah	40	90	...	11,107
Arizon	70	55	...	21,500
Washington	75	85	...	20,900
Colorado	65	45	...	21,336

DICE OPERATOR

- Focus on a subset of member values
- Replace dimension with a subset of values
- Dice operation often follows a slice operation

Location	U t a h	4 0	9 0	5 0	3 0
		M o n o L a s e r	I n k J e t	P h o t o	P o r t a b l e
		P r o d u c t			

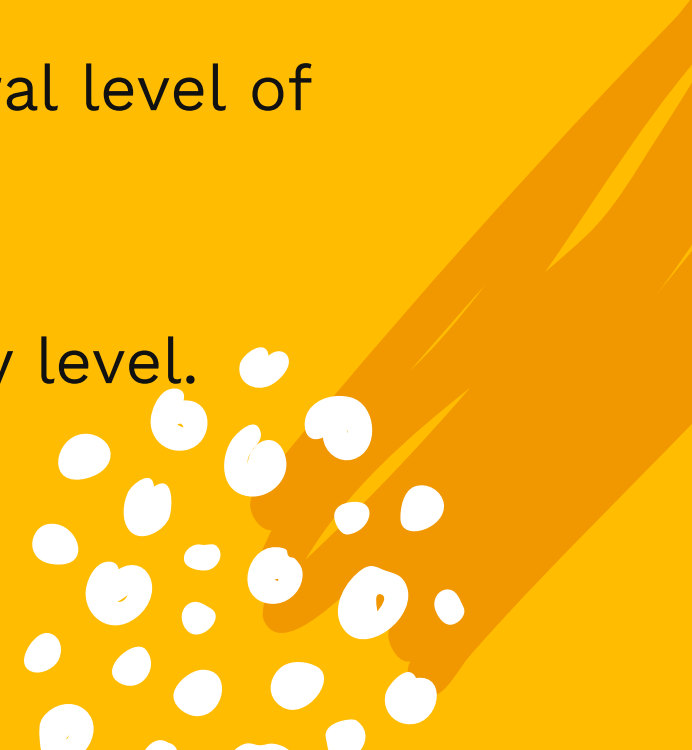
DRILL-DOWN

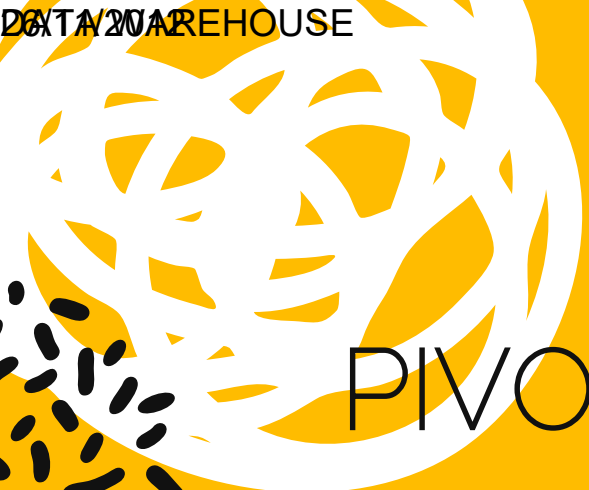
- navigate from a more general level to more specific.
- Obtain more detail about dimension.

Location	Product			
	Mono Laser	Ink Jet	Photo	Portable
California	80	110	60	25
Utah				
Salt Lake	20	20	10	15
Park City	5	30	10	5
Ogden	15	40	30	10
Arizon	70	55	60	35
Washington	75	85	45	45
Colorado	65	45	85	60



ROLL-UP/DRILL-UP

- x Remove detail from a dimension.
 - x Moving from a specific level to a more general level of a hierarchical dimension.
 - x eg: roll up sales data from daily to quarterly level.
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PIVOT

- Rearrange dimensions so that data cube can be presented in a visually appealing order.
 - Most typically used on data cube of more than two dimensions.
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References

- x Thomas Connolly and Carolyn Begg, Database Systems: A Practical Approach to Design, Implementation, and Management, 6th Edition, Pearson, 2015, ISBN: 978-01329432
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